

# Will Steinhardt

[wsteinh@gmail.com](mailto:wsteinh@gmail.com) | [Google Scholar](#) | [willsteinhardt.com](http://willsteinhardt.com)

## Appointments

UC Santa Cruz | Assistant Researcher  
UC Santa Cruz | Postdoctoral Scholar

2025-Current  
2020-2025

## Education

Harvard University | Ph.D. in Earth & Planetary Sciences  
Caltech | B.S. in Geophysics

Feb 2020  
2011

## Honors and Awards

Jason Morgan Early Career Award | AGU  
Koret Scholar Mentoring Award | UCSC  
Outstanding Student Presentation Award | AGU  
Shaler Teaching Award | Harvard Earth & Planetary Sciences  
Certificate of Distinction in Teaching | Harvard Bok Center  
NASA Earth and Space Science Fellowship | NASA (declined)  
Fritz Burns Prize | Caltech Division of Geologic & Planetary Sciences  
Student Life and Masters Award | Caltech

2024  
2023  
2018  
2016  
Spring 2015, Spring 2016  
2013  
2011  
2011

## Publications

1. Steinhardt, W. & Brodsky, E. E. "Distinguishing Single and Linked Ruptures in the Laboratory and Nature". *Geophysical Research Letters* 53, e2025GL120688 (2026). ([LINK](#))
2. Steinhardt, W. & Brodsky, E. E. "Precursory Locking Precedes Slip Events on Laboratory Fault". *Geophysical Research Letters* 52, e2024GL112882 (2025). ([LINK](#))
3. **Steinhardt, W.** & Rubinstein, S. M. "Geometric rules for the annihilation dynamics of step lines on fracture fronts". *Phys. Rev. E* 107, 055003 (2023). ([LINK](#))
4. **Steinhardt, W.**, Dillavou, S., Agajanian, M., Rubinstein, S. M. & Brodsky, E. E. "Seismological Stress Drops for Confined Ruptures Are Invariant to Normal Stress". *Geophysical Research Letters* 50, e2022GL101366 (2023). ([LINK](#))
5. **Steinhardt, W.** & Rubinstein, S.M. "How Material Heterogeneity Creates Rough Fractures" *Phys. Rev. Lett.*, 129, 128001 (2022). ([LINK](#)) ([Chosen For Cover](#))
6. Solomatova, N.V., Jackson, J.M., Sturhahn, W., Wicks, J.K., Zhao, J., Toellner, T.S., Kalkan, B. and **Steinhardt, W.** "Equation of state and spin crossover of (Mg, Fe) O at high pressure, with implications for explaining topographic relief at the core-mantle boundary." *American Mineralogist*, 101(5), pp.1084-1093. (2016).
7. Bindi, L., Yao, N., Lin, C., Hollister, L.S., Andronicos, C.L., Distler, V.V., Eddy, M.P., Kostin, A., Kryachko, V., MacPherson, G.J., **Steinhardt, W.**, Yudovskaya M., Steinhardt P.J., "Decagonite,  $\text{Al}_{71}\text{Ni}_{24}\text{Fe}_5$ , a quasicrystal with decagonal symmetry from the Khatyrka CV3 carbonaceous chondrite." *American Mineralogist*, 100(10), pp.2340-2343. (2015).
8. Bindi, L., Yao, N., Lin, C., Hollister, L.S., Andronicos, C.L., Distler, V.V., Eddy, M.P., Kostin, A., Kryachko, V., MacPherson, G.J., **Steinhardt, W.**, Yudovskaya M., Steinhardt P.J., 2015. "Natural quasicrystal with decagonal symmetry." *Scientific Reports*, 5, p.9111. (2015).
9. Bindi, L., Yao, N., Lin, C., Hollister, L.S., MacPherson, G.J., Poirier, G.R., Andronicos, C.L., Distler, V.V., Eddy, M.P., Kostin, A., Kryachko, V., **Steinhardt, W.**, Yudovskaya M., Steinhardt P.J., "Steinhardtite, a new body-centered-cubic allotropic form of aluminum from the Khatyrka CV3 carbonaceous chondrite." *American Mineralogist*, 99(11-12), pp.2433-2436. (2014).
10. Hollister, L.S., Bindi, L., Yao, N., Poirier, G.R., Andronicos, C.L., MacPherson, G.J., Lin, C., Distler, V.V., Eddy, M.P., Kostin, A., Kryachko, V., **Steinhardt, W.**, Yudovskaya M., Eiler J.M., Guan Y., Clarke J.J., Steinhardt P.J., "Impact-induced shock and the formation of natural quasicrystals in the early solar system." *Nature Communications*, 5, p.4040. (2014).
11. MacPherson, G.J., Andronicos, C.L., Bindi, L., Distler, V.V., Eddy, M.P., Eiler, J.M., Guan, Y., Hollister, L.S., Kostin, A., Kryachko, V., **Steinhardt, W.**, Yudovskaya M., Steinhardt P.J., "Khatyrka, a new CV 3 find from the Koryak Mountains, Eastern Russia." *Meteoritics & Planetary Science*, 48(8), pp.1499-1514. (2013).
12. M. Obrić, Ž Ivezić, P. N. Best, R. H. Lupton, C. Tremonti, J. Brinchmann, M. A. Agüeros, G. R. Knapp, J. E. Gunn, C. M. Rockosi, D. Schlegel, D. Finkbeiner, M. Gaćša, V. Smolčić, S. F. Anderson, W. Voges, M. Jurić, R. J. Siverd, **Steinhardt, W.**, A. S. Jagoda, M. R. Blanton, D. P. Schneider; "Panchromatic properties of 99 000 galaxies detected by SDSS, and (some by) ROSAT, GALEX, 2MASS, IRAS, GB6, FIRST, NVSS and

WENSS surveys”, Monthly Notices of the Royal Astronomical Society, Volume 370, Issue 4, 1677–1698 (2006).

13. Sesar, B., Svilković, D., Ivezić, Ž., Lupton, R.H., Munn, J.A., Finkbeiner, D., **Steinhardt, W.**, Siverd, R., Johnston, D.E., Knapp, G.R. and Gunn, J.E., “Variable faint optical sources discovered by comparing the POSS and SDSS catalogs.” The Astronomical Journal, 131(6), p.2801. (2006).
14. Ivezić, Ž., Richards, G., Hall, P., Lupton, R., Jagoda, A., Knapp, G., Gunn, J., Strauss, M., Schlegel, D., **Steinhardt, W.** and Siverd, R., June. “Quasar Radio Dichotomy: Two Peaks, or not Two Peaks, that is the Question.” AGN Physics with the Sloan Digital Sky Survey (Vol. 311, p. 347). (2004).

## Invited Talks

|                                                                  |           |
|------------------------------------------------------------------|-----------|
| Gordon Research Conference (Rock Mechanics)                      | 8 / 2026  |
| Gordon Research Conference (Soft Matter)                         | 7 / 2026  |
| USGS Earthquake Science Seminar                                  | 4 / 2026  |
| KITP Soft Matter Geophysics Seminar                              | 1 / 2026  |
| Berkeley Earth Science Seminar                                   | 1 / 2025  |
| Cornell Civil and Environmental Engineering Seminar              | 12 / 2024 |
| Berkeley Seismo Lab Seminar                                      | 10 / 2024 |
| Princeton Center for Theoretical Physics: Fracture Across Fields | 5 / 2024  |
| Computational Infrastructure for Geodynamics Seminar             | 3 / 2023  |
| Brandeis MRSEC Seminar                                           | 10 / 2022 |
| MIT Geophysics Seminar                                           | 10 / 2022 |
| USGS Earthquake Science Center Seminar                           | 7 / 2022  |
| University of Washington Earth and Space Science Colloquium      | 1 / 2022  |
| UCSC IGPP Seminar                                                | 2 / 2020  |
| American Physical Society (March Meeting 2019)                   | 3 / 2019  |

## Teaching

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|-------------------------------------------------------------------------------------------------|--------------------|
| Teaching Fellow for <i>ENG-SCI 123: Introduction to Fluid Mechanics and Transport Processes</i> | Spr 2015, Spr 2016 |
| Teaching Assistant for <i>GE 1: Earth and Environment</i>                                       | Spr 2010           |

## Service and Outreach

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|-----------------------------------------------------------------------------------------------------------|----------------|
| Reviewer for <i>Scientific Reports</i>                                                                    |                |
| Reviewer for <i>Review of Scientific Instruments</i>                                                      |                |
| Reviewer for <i>Journal of Geophysical Research</i>                                                       |                |
| Organized UCSC IGPP Seminar                                                                               | 2020 - 2022    |
| Supervised undergraduate Shannon Iron on earthquake statistics research                                   | 2023 - Present |
| Supervised undergraduate Jaiden Zak on friction research                                                  | 2022 - 2023    |
| Supervised undergraduate Sydney Haith on interacting asperity research                                    | 2022 – 2024    |
| Supervised graduate student Caroline Martin on fracture surface analysis project                          | 2019 - 2019    |
| Supervised graduate student Rodrigo Telles on hydrogel heterogeneity project                              | 2018 - 2019    |
| Supervised undergraduate Aria Hamann on interfacial hydrogel fracture project through Harvard REU Program | 2014           |
| Led <i>Science Days</i> lab tours and demonstrations for middle school students                           | 2015, 2016     |